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DOCUMENT 3: MATH101 Past Exam

UNIVERSITY MATHEMATICS DEPARTMENT

MATH101: Calculus I — Final Examination (Fall 2025 Archive)

Time Allowed: 3 Hours

Total Marks: 100 Points

Instructions: Show all working clearly. Unsupported answers will receive minimal credit. Non-graphing standard scientific calculators allowed.

Question 1 (8 Points)

Evaluate the following limit without using L'Hopital's Rule:

limit as x approaches 4 of (sqrt(x) - 2) / (x - 4)

Question 2 (10 Points)

Using the formal definition of a derivative (limit of a difference quotient), calculate f'(x) for:

f(x) = 1 / (2x + 1)

Question 3 (10 Points)

Find dy/dx for each of the following functions:

* **(a)** [5 Points] y = ln(sin(4x) + x^3)
* **(b)** [5 Points] y = x^(cos(x)) (Use logarithmic differentiation)

Question 4 (8 Points)

Find the equation of the tangent line to the implicit curve x^2 * y + y^3 * x = 10 at the point (1, 2).

Question 5 (12 Points)

A spherical balloon is being inflated with gas at a constant rate of 100 cm^3/sec.

* **(a)** [8 Points] How fast is the radius of the balloon increasing when the diameter is 50 cm?
* **(b)** [4 Points] How fast is the surface area increasing at that same instant?

Question 6 (12 Points)

An open-top rectangular box with a square base is to be constructed with a volume of 108 cubic feet. Find the dimensions of the box that will minimize the total surface area of materials used.

Question 7 (10 Points)

Let f(x) = (x^2 - 1) / (x^2 - 4). Find:

* **(a)** [3 Points] All horizontal and vertical asymptotes.
* **(b)** [4 Points] Intervals of increase and decrease.

* **(c)** [3 Points] Local maximum and minimum values.

Question 8 (10 Points)

Evaluate the following indefinite integrals:

* **(a)** [5 Points] integral of x^2 / sqrt(x^3 + 1) dx
* **(b)** [5 Points] integral of (3*x^2 + 2x) * cos(x^3 + x^2) dx

Question 9 (15 Points)

Evaluate the following definite integrals:

* **(a)** [7 Points] integral from 0 to 1 of x^2 * e^(-x) dx *(Hint: Integration by Parts)*
* **(b)** [8 Points] integral from 1 to e of ((ln(x))^2) / x dx

Question 10 (5 Points)

Let g(x) = integral from 2 to x^3 of sqrt(1 + t^4) dt. Use Part 1 of the Fundamental Theorem of Calculus along with the Chain Rule to calculate g'(x).